

Leveraging Machine Learning for Early Detection of Coronary Heart Disease Risk

Mengozzi Federico, Pucci Valentina, Silvagni Leonardo

Neuro-X Section, EPFL

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Abstract—Cardiovascular diseases (CVDs) remain a leading cause of mortality globally, emphasizing the need for effective early detection. This study leverages machine learning (ML) to predict coronary heart disease (CHD) risk using data from the Behavioral Risk Factor Surveillance System (BRFSS) [1].

I. INTRODUCTION

Coronary heart disease (CHD) remains a major contributor to global mortality, especially among older populations. To mitigate its health and economic impacts, early and accurate risk prediction is crucial. Machine learning (ML) offers promising methods for this purpose.

This study focuses on developing an ML-based predictive model for CHD risk using the Behavioral Risk Factor Surveillance System (BRFSS) dataset [1]. Key contributions include data preprocessing strategies for addressing missing values and feature engineering to improve model performance. Our model employs logistic regression optimized with stochastic gradient descent (SGD) and ridge regularization.

II. DATA ANALYSIS

A. Dataset Description

The Behavioral Risk Factor Surveillance System (BRFSS) [1] is a large-scale Cell-Phone survey dataset that collects information on health behaviors, chronic conditions, and demographics to evaluate health risks, including coronary heart disease (CHD). However, the dataset is not clean and contains missing values, with 56.5% of features having at least 10% missing data.

Moreover, the dataset is highly imbalanced, with 299,160 samples in the majority class (False) compared to only 28,976 in the minority class (True). This imbalance can cause machine learning algorithms to favor the majority class due to its larger representation in the data.

B. Handling Missing Values

To streamline the dataset, we first removed features with over 10% of missing data. For the remaining features, we used a complete-case analysis, excluding rows with any blank value to create a baseline model that relies only on complete cases.

For the final model, we selectively reinstated high-importance features that were initially removed, handling missing values with customized imputation strategies tailored to feature-specific contexts. For instance, setting *PREGNANT* to 0 for male participants (rather than leaving

it blank) allowed us to distinguish conditionally blank responses from genuinely missing data, enhancing model accuracy.

We evaluated various imputation methods for empty values, including distribution-based and placeholder-based imputations, ultimately choosing median imputation due to its superior predictive performance. This refined approach allowed us to retain valuable information, bolstering both data integrity and model quality.

C. Feature Engineering

Following feature selection, we implemented several enhancement steps, which were evaluated using cross-validation analysis (III-E) to determine the optimal approach.

In our baseline, we manually scaled features, such as *HEIGHT*, *WEIGHT*, and *ALCDAY5*, to ensure unit consistency and corrected improperly coded entries (e.g., values like ‘88’ and ‘99’ indicating “refuse to answer” or “not sure”) to enhance data accuracy. We then normalized all features to a 0 to 1 range for consistency across variables.

We experimented with different transformations for categorical features to further refine our model. Initially, we reordered feature values based on relative risk calculated from the training set. Subsequently, we tested one-hot encoding to convert categorical features into binary variables, removing any artificial ordinal relationships. Based on the performance, one-hot encoding was selected as the final transformation method.

III. MODELS AND METHODS

A. Model selection

We selected Logistic Regression as the most suitable choice for this study due to its binary output, interpretability, and robustness. To optimize the model, we enhanced it with stochastic gradient descent (SGD) and ridge regularization. We opted for SGD over traditional gradient descent due to its efficiency with large datasets, faster convergence, and superior generalization.

We considered K-Nearest Neighbors (KNN) for model training and *Dataset Augmentation* (III-B), but found it unsuitable for high-dimensional data with weak spatial correlation. Support Vector Machines (SVM) were also excluded, as their reliance on linear separability and Euclidean distances is less effective for predominantly categorical data.

We also explored Ridge Regression with SGD, but it demonstrated inferior performance compared to our optimized Logistic Regression model, aligning with our theoretical expectations.

B. Dataset Augmentation

To address class imbalance, we experimented with various resampling techniques, including random undersampling, random oversampling, combined over and under-sampling, SMOTE[2], ADASYN[3], and Tomek links[4]. Ultimately, using the cross-validation technique (III-E), we selected random undersampling with the optimal ratio between the minority and majority classes of 0.33 as the most performing method. We used it in cross-validation on the training batches while, instead, we kept validation batches unbalanced as in the real test dataset.

C. Feature expansion

To improve model performance, we applied polynomial expansion to non-categorical features. Degrees ranging from 2 to 5 were tested using cross-validation (III-E), with degree 3 providing the best results. In contrast, categorical features were expanded through one-hot encoding.

D. Threshold selection

To further fine-tune our model, we explored adjusting the classification threshold for the logistic regression. Using the established framework, we selected the optimal threshold based on the data presented in Figure III-D. As shown in the plot, we chose the actual maximum F1 score, achieved with a threshold of 0.4517 , rather than relying on the maximum of the quadratic interpolation, as this yielded superior results.

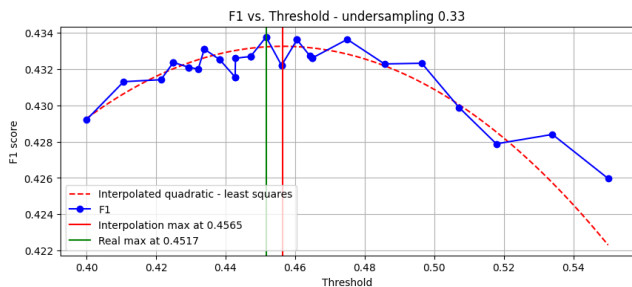


Figure 1. F1 score plot against Threshold for logistic regression

E. K-Fold Cross Validation

To validate our preprocessing approach, we applied 5-fold cross-validation within the training dataset. To choose the optimal hyperparameters, we considered various metrics, pondering more the F1 score for its ability to better represent model performance in the case of unbalanced datasets. Considering only accuracy for this dataset would not have been indicative, because accuracy would spike to 91.17% , which is the proportion of the majority class in the dataset, when a constant prediction of the majority class is made.

The optimal hyperparameters for each classifier are summarized in Table I, while each step’s impact on model performance is detailed Table II, where we evaluated improvements based on F1-score and accuracy. Notably, one-hot encoding and the undersampling technique emerged as the most effective strategies.

Table I
OPTIMAL HYPERPARAMETERS FOR LOGISTIC REGRESSION

Hyperparameter	Optimal Value
λ (Reg)	1e-6
Polynomial Degree	3
Learning Rate (Gamma SGD)	0.01
Number of Iterations	1500
Number of Batches	200
Batch Size	50

Table II
PERFORMANCE METRICS AFTER EACH FEATURE ENGINEERING STEP

Preprocessing Step	Accuracy	F1-Score
Baseline, complete cases	0.912 ± 0.004	0.262 ± 0.002
Median imputation	0.913 ± 0.003	0.280 ± 0.003
One-Hot Encoding	0.912 ± 0.003	0.335 ± 0.002
Undersampling (ratio=0.33)	0.872 ± 0.004	0.432 ± 0.004
Threshold selection	0.872 ± 0.002	0.434 ± 0.002

IV. RESULTS

Our final model achieved the results presented in Table III for cross-validation. Our final model achieved an *Accuracy* of 0.874 and an *F1-Score* of 0.440 on the AiCrowd platform challenge.

Table III
FINAL RESULTS GIVEN BY OUR MODEL

Accuracy	F1-Score	Recall	Precision	Specificity
0.872	0.434	0.558	0.356	0.902

V. CONCLUSIONS

This study demonstrates that targeted preprocessing and feature engineering significantly enhance prediction accuracy in assessing coronary heart disease (CHD) risk. While our logistic regression model showed promising results, there is potential for further improvement.

Future research could investigate advanced feature engineering techniques or explore alternative models better suited to this medical dataset.

One promising direction is to leverage the interpretability of the logistic regression model to automatically identify and select the most significant features, discarding those of lesser relevance.

Additionally, weighted gradient descent (GD) can help address class imbalance in this dataset. Given the limited number of True labels compared to False labels, applying a higher weight to the prediction error for positive cases can prevent the model from being biased toward a low recall—a common issue when the model tends to favor the majority class.

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